

CRP HEL and Wetland Screening Tool

(Prototype)

Methodology, Formula Reference & Data Sources

Based on: USDA NRCS SSURGO Database, NRCS FOTG R-Factors, NOAA Climate Data Online API, USGS 3DEP Elevation Data, NLCD Land Cover Classification, NHD/NWI Wetland Database

Prepared by: Shilpi Gupta | AI Product Strategist & Technical PM

Contact: shilpigupta30@gmail.com | **Date:** May 2026 | **Version:** 5.1

Status: Results are indicative only — not an official RUSLE2 or HEL determination

WETLAND DETERMINATION METHODOLOGY (NEW)

Tool now includes comprehensive 4-indicator wetland assessment per Federal Interagency Wetlands Delineation Manual:

1. Hydric Soils Detection (SSURGO hydricating)

- **Source:** SSURGO hydricating field (NRCS soil property)
- **Accuracy:** $\pm 5\%$ (established soil classification)
- **Signal:** Soils formed under prolonged saturation conditions
- **Implementation:** Queries SDA for hydricating='yes' at field location

2. Hydrophytic Vegetation Detection (NLCD Land Cover)

- **Source:** NLCD 2019 Land Cover (MRLC WMS GetFeatureInfo API)
- **Classes:** Class 90 (Woody Wetlands) & Class 95 (Emergent Herbaceous Wetlands)
- **Accuracy:** $\pm 5\%$ (30m resolution Landsat-derived classification)
- **Signal:** Presence of wetland vegetation types
- **Implementation:** Point-specific NLCD query returns vegetation class with confidence

3. Wetland Hydrology Detection (Water Table Estimation)

- **Source:** SSURGO drainage class (proxy for water table depth)
- **Method:** NRCS hydrologic correlation with drainage classes
 - Very Poorly Drained $\rightarrow$ 10 cm water table (Strong signal)
 - Poorly Drained $\rightarrow$ 25 cm water table (Strong signal)
 - Somewhat Poorly Drained $\rightarrow$ 45 cm water table (Possible signal)
 - Well/Moderately Well Drained $\rightarrow$ >100 cm (No signal)
- **Accuracy:** $\pm 20\%$ (established NRCS soil morphology standards)
- **Implementation:** Dominant component drainage class determines water table category

4. Proximity to Water Body Detection (NHD/NWI Database)

- **Source:** FWS National Wetlands Inventory (NWI) via WMS GetFeatureInfo
- **Coverage:** Mapped wetland polygons across all US states
- **Classes:** Cowardin Classification (P=Palustrine, R=Riverine, L=Lacustrine, E=Estuarine, M=Marine)
- **Accuracy:** $\pm 5\%$ (official USGS/FWS wetland delineation)
- **Confidence Levels:**
 - Strong: P/R/L systems (freshwater palustrine, riverine, lacustrine)
 - Possible: E/M systems (estuarine, marine)
- **Implementation:** 5km search radius from field center using NWI WMS
- **Details Provided:** Wetland type, NWI Cowardin code, hydrology signal strength

Determination Confidence Scoring:

- **High Confidence (3+ indicators):** Strong candidate for wetland restoration (CP23, CP28)
- **Medium Confidence (2 indicators):** Possible wetland restoration opportunity - field verification recommended
- **Low Confidence (1 indicator):** Potential wetland - further investigation needed
- **No Indicators:** Does not appear to be wetland-forming soil

Important: Official wetland determination requires NRCS field visit and certification by NRCS conservationist per 7 CFR §§ 12.20–12.30

UI DISPLAY ENHANCEMENTS

Detailed Wetland Indicators Table (Conservationist View)

The tool now displays all 4 indicators with detailed evidence from actual API queries:

Example Output (Atchafalaya Basin, LA):

Indicator	Detected	Evidence	Confidence
Hydric Soils	Yes	SSURGO hydricrating present	High
Hydrophytic Vegetation	Yes	NLCD Class 90: Woody Wetlands	High
Wetland Hydrology	Yes	Poorly drained (25cm estim.)	High
Proximity to Water Body	Yes	NWI: PFO1/2F - Strong signal	High

Benefits of Enhanced Display:

- Shows actual NLCD class numbers and vegetation types (not just "yes/no")
- Displays NWI wetland codes and hydrology signal strengths
- Provides water table depth estimates derived from drainage class
- Links evidence directly to official data sources (SSURGO, NLCD, NWI)
- Enables conservationists to verify data before field visit